



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

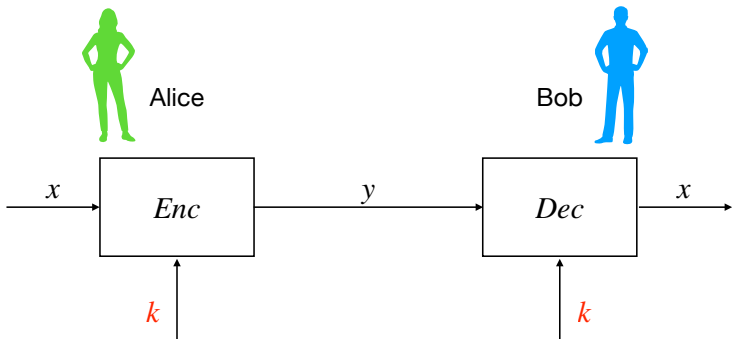
Introduction to Cryptography and Security

Introduction to Public-key Cryptography

Outline

- 1 Big issue with using symmetric algorithms
- 2 Asymmetric Cryptography
- 3 Essential Number Theory for Public-Key Algorithms

Symmetric Cryptography



- The same secret key k is used for encryption and decryption.
- The encryption and decryption function are very similar (in the case of DES they are essentially identical).

Symmetric Cryptography Revisited

- Modern symmetric algorithms such as AES or 3DES are very secure, fast and are in widespread use.
- But the key must be established between Alice and Bob using a secure channel.

Key management

- n users.
- Storing mutual secret key is difficult.

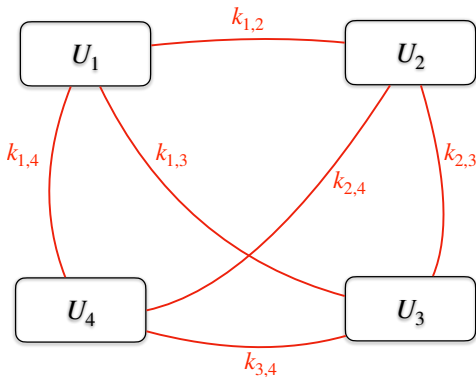


Figure: $O(n)$ keys per user

No Protection Against Cheating

by Alice or Bob

- Alice and Bob have the same capabilities, since they possess the same key.
- For example, Alice can claim that she never ordered online TV from Bob (he could make the order himself).
- Preventing this is called **nonrepudiation** and can be achieved with asymmetric cryptography.

Outline

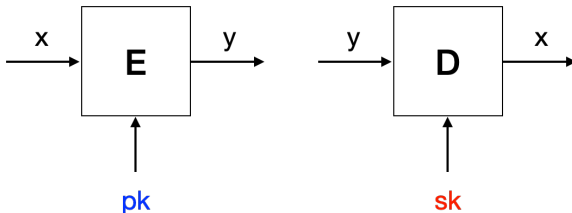
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Principles of Asymmetric Cryptography

Whitfield Diffie, Martin Hellman, and Ralph Merkle – 1976



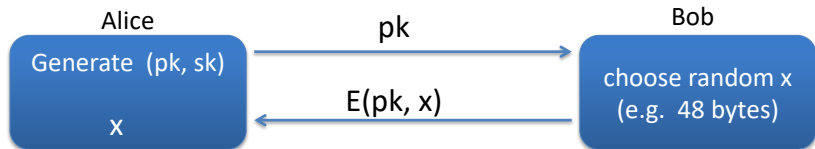
Public key encryption



Bob generates $k = (pk, sk)$ and gives pk to Alice.

Application: Session setup

for now, only eavesdropping security



Non-interactive applications: : (e.g. Email)

- Bob sends email to Alice encrypted using pk_{alice}
- **Note:** Bob needs pk_{alice} (public key management)

Public key encryption

Definition

A **Public key encryption system** is a triple of algorithms

$$(G, E, D)$$

where:

$G()$: **randomized** algorithm output a key pair (pk, sk)

$E(pk, m)$: **randomized** algorithm that takes $m \in M$ and output $c \in C$

$D(sk, c)$: **deterministic** algorithm that takes $c \in C$ and output $m \in M$ or $\perp$

Consistency: For all (pk, sk) output by G :

$$\forall m \in M : D(sk, E(pk, m)) = m.$$

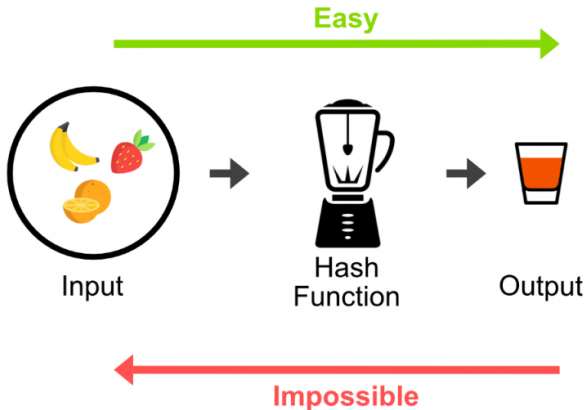
One-way function

The public-key algorithms are all built from one common principle, the *one-way function*.

- 1 $y = f(x)$ is computationally *easy*,
- 2 $x = f^{-1}(y)$ is computationally *infeasible*.



One-way function



Source: <https://computersciencewiki.org>

Public-Key Algorithm Families

- ***Integer Factorization Scheme***: The most prominent representative of this algorithm family is RSA.
- ***Discrete Logarithm Scheme***: For example, the Diffie–Hellman key exchange, Elgamal encryption or the Digital Signature Algorithm (DSA).
- ***Elliptic Curve Schemes***: For example, Elliptic Curve Diffie–Hellman key exchange (ECDH) and the Elliptic Curve Digital Signature Algorithm (ECDSA).

Comparison

Symmetric key size (bits)	Elliptic Curve	RSA or Diffie-Hellman
80	160	1024
112	224	2048
128	256	3072
192	384	7680
256	521	15360

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Definition

- The greatest common divisor (gcd) of two integers a and b is an integer d such that:

$$d \mid a \quad \text{and} \quad d \mid b.$$

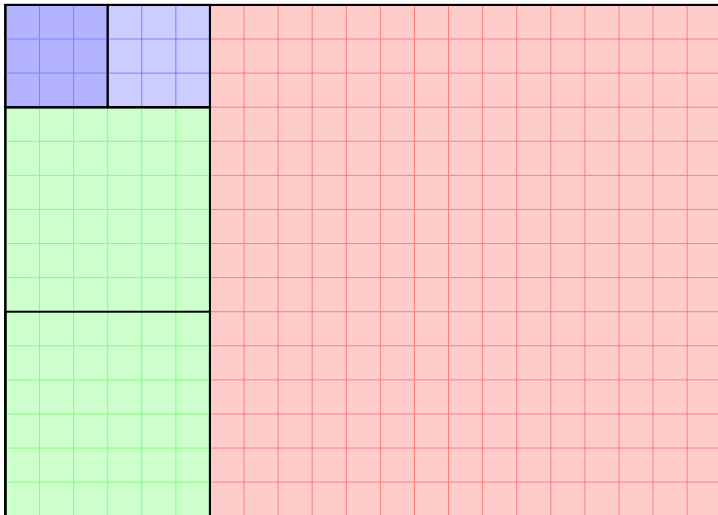
- We denote $\gcd(a, b)$ as the **largest** common divisor of a and b .

Example

- $\gcd(12, 18) = 6$ because $6 \mid 12$ and $6 \mid 18$, and there is no larger number with this property.
- $\gcd(748, 2014) = 44$ because

the divisors of 748 = {1, 2, 4, 11, 17, 22, 34, 44, 68, 187, 374, 748},
the divisors of 2024 = {1, 2, 4, 8, 11, 22, 23, 44, 46, 88, 92, 184, 253,
506, 1012, 2024}.

$$\gcd(21, 15) = \gcd(15, 6) = \gcd(6, 3)$$



Theorem (Euclidean Algorithm)

Let a, b be two positive integers with $a \geq b$. The following algorithm computes $\gcd(a, b)$ in a finite number of steps.

- 1 Set $r_0 = a$ and $r_1 = b$.
- 2 Set $i = 1$.
- 3 Divide r_{i-1} by r_i , we get

$$r_{i-1} = r_i \cdot q_i + r_{i+1} \quad \text{with} \quad 0 \leq r_{i+1} < r_i.$$

- 4 If $r_{i+1} = 0$, then

$$r_i = \gcd(a, b)$$

and the algorithm terminates.

- 5 Otherwise, $r_{i+1} > 0$, then set $i = i + 1$ and go back to Step 3.

Theorem

The division (Step 3) of the Euclidean Algorithm is performed at most

$$\log_2(b) + 2 \quad \text{times.}$$

Euclidean Algorithm (recursive form)

```
EUCLID( $a, b$ )  
  if  $b == 0$   
    return  $a$   
  else  
    return EUCLID( $b, a \bmod b$ )
```

Extended Euclidean Algorithm

- The Euclidean Algorithm can be extended to find additional information.
- Specifically, we extend the algorithm to also compute coefficients x, y that satisfy

$$d = \gcd(a, b) = ax + by.$$

- The coefficients x, y can be negative or zero. These coefficients will be useful later when finding the multiplicative inverse in modular arithmetic.

Extended Euclidean Algorithm

- *Input* : Pair of positive integers (a, b)
- *Output*: Triple (d, x, y) satisfying

$$d = \gcd(a, b) = ax + by.$$

EXTENDED-EUCLID(a, b)

if $b == 0$

return $(a, 1, 0)$

else

$(d', x', y') = \text{EXTENDED-EUCLID}(b, a \bmod b)$

$(d, x, y) = (d', y', x' - \lfloor a/b \rfloor y')$

return (d, x, y)

Correctness of the Algorithm

- The algorithm finds (d, x, y) satisfying

$$d = \gcd(a, b) = ax + by$$

- If $b = 0$, then

$$d = a = a \cdot 1 + b \cdot 0.$$

- If $b \neq 0$, the EXTENDED-EUCLID algorithm will compute (d', x', y') satisfying

$$\begin{aligned} d' &= d = \gcd(b, a \bmod b) \\ &= bx' + (a \bmod b)y' \end{aligned}$$

- And then

$$\begin{aligned} d &= b'x' + (a - b\lfloor a/b \rfloor)y' \\ &= ay' + b(x' - \lfloor a/b \rfloor y') \end{aligned}$$

Example

a	b	$\lfloor a/b \rfloor$	d	x	y
99	78	1	3	-11	14
78	21	3	3	3	-11
21	15	1	3	-2	3
15	6	2	3	1	-2
6	3	2	3	0	1
3	0	-	3	1	0

- Each row of the table describes a recursive level: the input values a and b , the computed value $\lfloor a/b \rfloor$, and the returned values d, x, y .
- The triple d, x, y returned becomes the triple d', x', y' of the next level from the formula

$$x = y'$$

$$y = x' - \lfloor a/b \rfloor y'$$

Exercise

Calculate the value

$$(d, x, y) = \text{EXTENDED-EUCLID}(899, 493).$$

Calculating the Inverse

- Given $n > 1$, if $\gcd(a, n) = 1$ then we have

$$\gcd(a, n) = 1 = ax + ny$$

- Thus, $ax = 1 \pmod{n}$. That is

$$x = a^{-1} \pmod{n}$$

Calculating the Modular Inverse

- *Input* : Number $n > 0$ and number $a \in \mathbb{Z}_n$ such that $\gcd(a, n) = 1$
- *Output*: Number b satisfying $a \cdot b = 1 \pmod n$.

MOD-INV (a, n)

$(d, x, y) = \text{EXTENDED-EUCLID}(a, n)$

$b = x \pmod n$

return b

Example: Calculate $5^{-1} \bmod 12$

a	b	$\lfloor a/b \rfloor$	d	x	y
5	12	0	1	5	-2
12	5	2	1	-2	5
5	2	2	1	1	-2
2	1	2	1	0	1
1	0	-	1	1	0

Definition (Euler's Phi Function)

The number of integers in $\mathbb{Z}_m$ relatively prime to m is denoted by $\Phi(m)$.

Example

Let $m = 6$. The associated set is $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$.

$$\gcd(0, 6) = 6$$

$$\gcd(1, 6) = 1 \quad (\text{relatively prime})$$

$$\gcd(2, 6) = 2$$

$$\gcd(3, 6) = 3$$

$$\gcd(4, 6) = 2$$

$$\gcd(5, 6) = 1 \quad (\text{relatively prime})$$

There are two numbers that are relatively prime to 6, thus $\Phi(6) = 2$.

Example

Let $m = 5$. The associated set is $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$.

$$\gcd(0, 5) = 5$$

$$\gcd(1, 5) = 1 \quad (\text{relatively prime})$$

$$\gcd(2, 5) = 1 \quad (\text{relatively prime})$$

$$\gcd(3, 5) = 1 \quad (\text{relatively prime})$$

$$\gcd(4, 5) = 1 \quad (\text{relatively prime})$$

There are four numbers that are relatively prime to 5, thus $\Phi(5) = 4$.

Theorem

Let m have the following canonical factorization

$$m = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_n^{e_n}$$

where the p_i are distinct prime numbers and e_i are positive integers, then

$$\Phi(m) = \prod_{i=1}^n (p_i^{e_i} - p_i^{e_i-1})$$

Example

Let $m = 240$. The canonical factorization is

$$m = 240 = 2^4 \cdot 3 \cdot 5.$$

Euler's phi function is:

$$\Phi(240) = (2^4 - 2^3)(3 - 1)(5 - 1) = 8 \cdot 2 \cdot 4 = 64.$$

Thus, 64 integers in $\{0, 1, \dots, 239\}$ are coprime to 240.

Theorem (Fermat's Little Theorem)

Let a be an integer and p a prime, then:

$$a^{p-1} \equiv 1 \pmod{p}$$

This implies the multiplicative inverse of a modulo p is:

$$a^{-1} \equiv a^{p-2} \pmod{p}$$

Note: This holds only if p is a prime.

Example

Let $p = 7$ and $a = 2$. We can compute the inverse of a as:

$$a^{p-2} = 2^5 = 32 \equiv 4 \pmod{7}.$$

This is easy to verify:

$$2 \cdot 4 \equiv 1 \pmod{7}.$$

A generalization of Fermat's Little Theorem to any integer moduli, i.e., moduli that are not necessarily primes, is Euler's theorem.

Theorem (Euler's Theorem)

Let a and m be integers with $\gcd(a, m) = 1$, then:

$$a^{\Phi(m)} \equiv 1 \pmod{m}$$

Fermat's Little Theorem is a special case of Euler's theorem. For prime p , $\Phi(p) = p - 1$:

$$a^{p-1} \equiv 1 \pmod{p}$$

Example

Let $m = 12$ and $a = 5$. Compute Euler's phi function:

$$\Phi(12) = (4 - 2)(3 - 1) = 4$$

Verify Euler's theorem:

$$5^4 = 625 \equiv 1 \pmod{12}$$



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SOICT

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Thank you!

